

ST309 Group Project Report

Quantifying the Different Factors that Affect London House Prices

Candidate Number	Contribution
71149	1/3
72381	1/3
68540	1/3

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Introduction

Property prices in London display great variation across space and over time. Even within the same neighbourhood, properties of similar appearance and size may be bought and sold at very different prices. Understanding the factors affecting these price differences is the central problem tackled in this project. Particularly, this study investigates which property characteristics are most important in explaining varying residential property transaction prices across London. Using detailed transaction level data, the project focuses on how structural features of properties, tenure, type as well as geographic location, jointly affect market prices.

Knowing what drives residential property prices in London matters for both economic research and policy decisions. London is one of the world's most expensive housing markets, and rising prices have raised concerns about affordability, inequality, and access to homes. For most households, housing is their biggest asset and expense. For policymakers, property values shape discussions about planning, housing supply, and environmental rules. Particularly, London's Mayor Sadiq Khan has been outspoken about his desire to increase affordable housing in London by building on the green belt (London City Hall, 2025). Measuring how different features affect prices helps evaluate policies like this, understand price differences across the city, and gauge the impact of changes in building standards or housing types.

The problem can naturally be formulated as a data analytic problem because residential transactions generate large volumes of structured, quantitative data. Each transaction is associated with a historic sale price and a current sales estimate as well as a range of measurable attributes, including the size of the property, the number of rooms, the type of dwelling, tenure (leasehold vs freehold), and geographic location. These variables provide a basis for modelling house prices as a function of multiple potentially correlated explanatory factors. The complexity of the housing market, which involves non-linear relationships and interactions between characteristics, further motivates the use of modern statistical learning techniques alongside classical regression methods.

The primary goal of this study is to identify and quantify the relative importance of key observable drivers of housing prices in London. In particular, the project aims to evaluate how strongly structural characteristics such as floor area and number of bedrooms explain variation in transaction prices, and to assess the additional contribution of tenure and spatial location and proximity to hospitals, schools and tube stations (Jones, 2025). A further objective is to compare the predictive performance of alternative modelling approaches, including linear and non-linear methods, in order to determine which models provide the most accurate and interpretable representation of price formation in the London housing market.

The analysis of housing prices has a long tradition in the subject area, most notably through the hedonic pricing framework (ONS, 2025). In this literature, property prices are decomposed into implicit prices for individual attributes, allowing researchers to value specific characteristics such as size, quality or environmental factors. Hedonic models have been widely applied in urban economics and real estate research to study price gradients, neighbourhood effects and the valuation of amenities. More recently, researchers have increasingly adopted machine learning approaches to overcome limitations of purely linear specifications and to capture complex patterns in large housing datasets.

Previous attempts using data analysis have demonstrated that property characteristics and location are consistently among the strongest predictors of housing prices, with non-linear methods such as tree based models often outperforming traditional regression in predictive accuracy. Studies using transaction level data have shown that variables such as floor area, property type and spatial accessibility account for a substantial share of observed price variation, but also highlight the presence of strong interactions between attributes and location. However, many existing studies rely on relatively coarse property information or focus on limited geographic areas. By exploiting a large London-specific dataset containing rich structural and spatial information, this project seeks to contribute to this literature by providing a systematic comparison of classical and modern statistical learning approaches for explaining and predicting residential property prices in one of the most complex housing markets in the UK.

Data Cleaning and Preparation

The raw housing dataset (from Kaggle) initially contained around 418,000 observations, but many of these were repeated records of the same property. Our historical sales data ran from 1995 to 2024 and implied that the average property in the dataset was sold about 3 times in 30 years. We chose to just include the latest entry of each property by tracking its latest sale. We made this decision because we did not have data on renovations in a property and tracking the appreciation of a certain property without data on underlying changes was not particularly relevant to the question we were seeking to answer. After deduplication, this reduced to approximately 137,000 unique properties.

From this set, further cleaning was required to ensure a consistent modelling sample. Firstly, we removed all data which did not have key information that we needed including a current property price estimate and the tenure of the property. A small number of observations, around 80 properties, were also excluded as they were not located within a London borough.

We conducted further data cleaning and created 2 datasets. In the first dataset (which we used for our initial data exploration), all properties which lacked even 1 of essential structural information, specifically

bedrooms, bathrooms, living rooms and floor area were removed, following the approach adopted in previous work on linked UK housing and EPC data. Similarly, Chi et al. (2021) remove transactions where essential property attributes such as total floor area and number of habitable rooms are missing or clearly erroneous prior to analysis, in order to ensure a consistent and economically meaningful modelling sample. However, in the second dataset (used for our regression models), we did not remove the entries which were missing only 1 of these values. Instead, we tried to predict them using the other data we had. For example, if we had the 3 of the 4 data points namely the bedrooms, square footage and living room data for an apartment and if we had data on similar apartments in the area, we would predict the likely number of bathrooms for this apartment and then include this observation in our dataset.

The first method led to 95,000 observations whereas the second provided us with an additional 13,000 observations bringing the total to approximately 108,000. We tested our regression models on both these datasets but found practically the same results from both. Finally, we winsorised (by removing the top and bottom 1% of observations) and deleted high influence observations, resulting in a clean and economically meaningful sample suitable for modelling.

We also deliberately chose to exclude the rent estimate as a regressor because it is almost mechanically derived from the sale estimate price, the dependent variable. Including it would introduce endogeneity by effectively conditioning on a function of the outcome, inflating fit and rendering coefficients uninterpretable. A similar variable-selection decision was adopted by Ogundeji (2024), who removed price per square area from the set of explanatory variables because it is highly correlated with the house price, in order to avoid redundancy and multicollinearity.

Skewed variables such as local crime intensity and number of shops within 1 km were log transformed to capture proportional effects and reduce the influence of extreme values. Data on a few other important aspects was not available to us, including for Energy Ratings, which we would have liked to use in our regressions but data was missing for many observations and an Energy Rating is a difficult item to impute from other data since it depends very much on the particular house. Age of the building and data on maintenance of the property was also missing from our dataset.

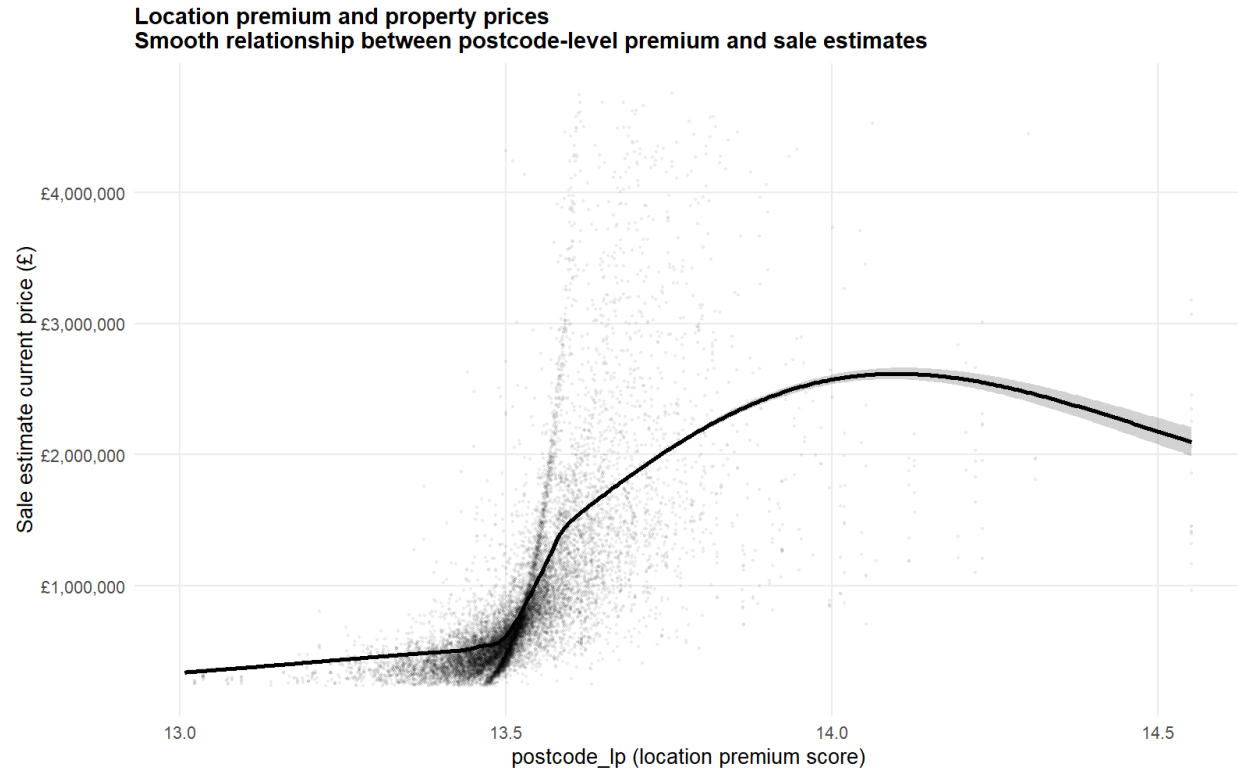
A simplifying assumption that we made was to classify each property by borough and to use the crime data at the borough level to signify the rate of crime for the houses. Gibbons (2004) for example, does not match crimes to individual properties, but instead constructs measures of local crime density in the neighbourhood of each property in order to capture expected crime exposure. We also chose to equate the crime data in the City of London to the neighbouring district of Westminster as our data source (the

Metropolitan Police District) did not have crime data for the City of London as the City has its own police force called the City of London Police. We also removed houses which were “shared” and allowed for partial ownership as these had unique circumstances and unrealistic sales estimates, however, there were very few observations like these in our dataset (less than 1000). This practice is consistent with the way UK price statistics treat atypical sales: official price datasets only include sales at full market value to avoid biasing price measures (Office for National Statistics, 2019).

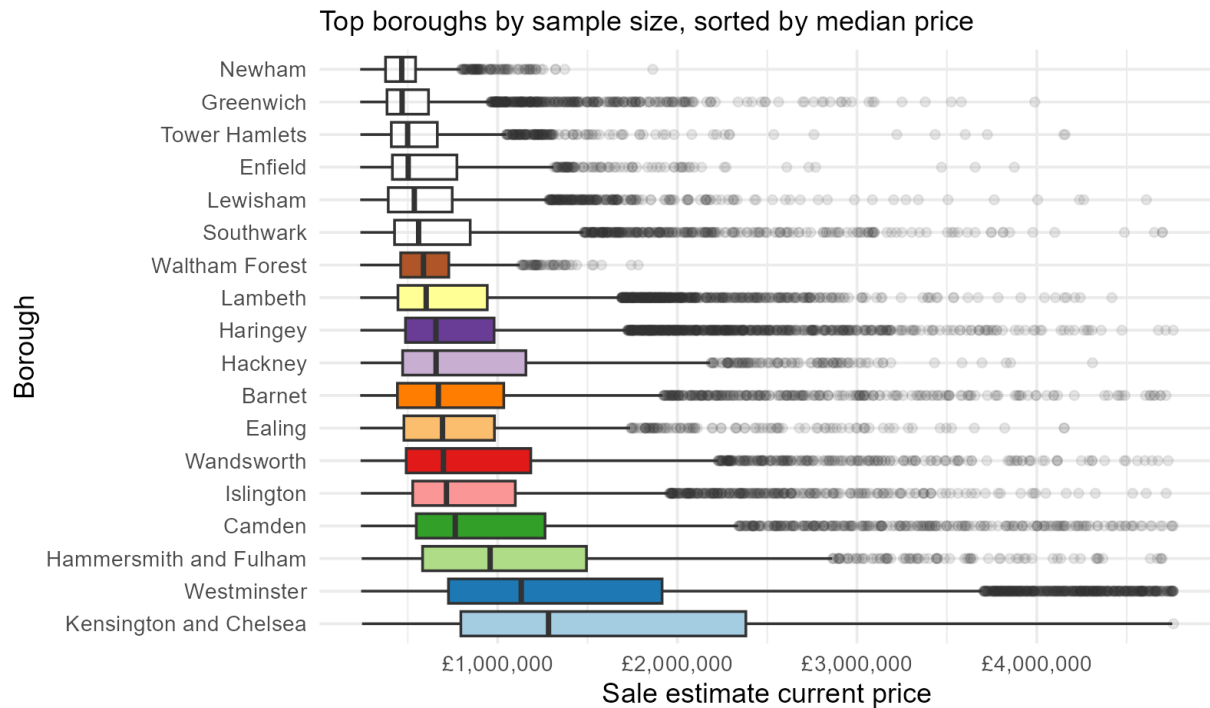
Initial Data Exploration & Model Set-Up

A key feature of the model is the treatment of historical transaction prices. Each property’s past sale price is adjusted into real December 2025 pounds using the UK CPIH index. Historical prices are matched to monthly CPIH values based on transaction dates, and an inflation factor is applied to ensure all prices are expressed in comparable real terms. The log of the inflation-adjusted historical price is retained as an informative signal, but is not directly included in the regression to avoid leaking future information into the model.

Instead, historical prices are used to construct a postcode-level location premium through empirical Bayes shrinkage. For each postcode, we compute the average log inflation-adjusted historical price and regularise it toward the London-wide mean using a smoothing parameter. Postcodes with many transactions receive estimates close to their raw average, while those with few observations are pulled toward the global mean, ensuring that postcodes with limited sales do not generate unstable estimates. By construction, the premium reflects revealed preferences from historical transactions (adjusted to December 2025 pounds) rather than current valuations, reducing concerns about mechanical correlation with the dependent variable. This variable captures persistent spatial desirability driven by factors such as reputation, connectivity, and long-run demand that are difficult to observe directly; helping us capture the premium associated with areas like Knightsbridge and Mayfair which may come from buyer sentiment more than easily measurable factors like square footage. It also prevents us from having to control for time fixed effects.



The figure above shows that the postcode location premium factor has a strongly non linear relationship with prices. This supports our later use of non-linear models to estimate house prices given that location premium is a key aspect. Around the median of the distribution, the slope is the steepest, which explains the large location effects estimated by the non linear models. The negative slope at the extreme upper end of the graph likely reflects limited observations and greater heterogeneity among ultra-prime postcodes, which is consistent with greater noise in that part of the graph. Another possible explanation is a change in demand dynamics among these prime areas: our premium estimate is built with historical prices and perhaps those areas with properties that sold for the highest prices in the past are no longer the most desirable locations and have relatively lower sales estimates currently.



Our model has the logarithm of the current sale price estimate as the dependent variable. Modelling prices in logs stabilises variance, reduces the influence of extreme values, and allows coefficients to be interpreted approximately as percentage effects. The core housing characteristics included are bedrooms, bathrooms, floor area, which were left in levels to support interpretability. Another important predictor was tenure, which was encoded as a binary indicator distinguishing freehold from leasehold properties.

The model also incorporates neighbourhood characteristics derived from OpenStreetMap and crime data. These include distances to and densities of tube stations, parks, schools, shops, hospitals, and police stations, as well as recent crime intensity. All such variables are also log-transformed, reflecting diminishing marginal effects. For example, reducing distance to a tube station from several kilometres to one kilometre is far more valuable than reducing it by a few additional metres once already nearby.

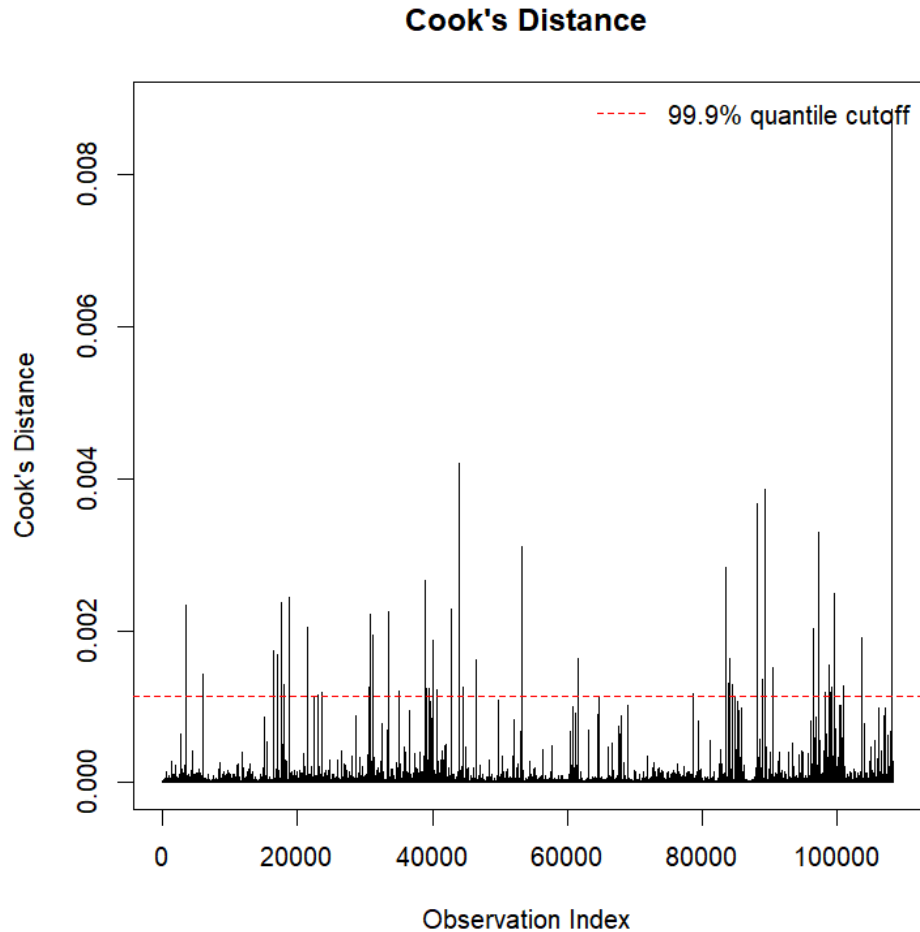
Before modelling, the data are split into training and test samples (80-20). To ensure the reliability of our regression estimates, we first examine multicollinearity among predictors and identify potentially influential observations.

Variance Inflation Factors (VIF)

Variable	GVIF	df	$GVIF^{1/(2\ df)}$
Bedrooms	3.549	1	1.884
Bathrooms	1.917	1	1.385
Floor Area	4.264	1	2.065
Living Rooms	1.778	1	1.333
Tenure	2.120	1	1.460
Borough (FE)	1.361	28	1.005

Notes: Reported values are generalized variance inflation factors (GVIF). For multi-level factors (e.g., borough fixed effects), the adjusted measure $GVIF^{1/(2\ df)}$ is reported to allow comparability across variables.

The variance inflation factors indicate that multicollinearity is not a concern in our dataset. Floor area exhibits the highest inflation ($GVIF^{1/(2df)} = 2.065$), suggesting modest correlation with other size-related variables, though this remains well below conventional thresholds. Bedrooms show similarly acceptable inflation ($GVIF^{1/(2df)} = 1.884$), likely reflecting its natural relationship with overall dwelling size. The remaining variables; Tenure, Bathrooms, and Living Rooms, demonstrate minimal multicollinearity, indicating each contributes independent information to the model. Borough fixed effects exhibit virtually no inflation, confirming that geographic controls are orthogonal to other predictors and do not introduce collinearity issues. Overall, all variance inflation factors fall comfortably below the threshold of 5, with most below 2, suggesting that standard errors and coefficient estimates are stable and reliable for statistical inference.



The distribution shows most Cook's Distance values concentrated near zero, with occasional spikes scattered throughout the dataset, and several observations approaching or marginally exceeding the 99.9% quantile threshold (red dashed line), suggesting these cases exert disproportionate influence on the coefficient estimates. Given that the vast majority of observations cluster near zero, a typical $4/n$ criterion would risk flagging and removing even minor deviations from the observations set, that represent legitimate variation rather than problematic outliers. Such aggressive trimming could paradoxically reduce model robustness by eliminating data points that contribute to understanding the full range of relationships in the population. The 99.9% threshold instead targets only the most extreme influential cases that could genuinely distort coefficient estimates, while preserving the broader distribution of observations that enhances model generalisability and statistical power.

Model Results

All models are estimated on the same winsorised sample and share an identical train–test split, ensuring that differences in performance reflect modelling choices rather than differences in underlying data.

Cross-validation RMSE is computed on the training set only using 5-fold CV, while test RMSE and R^2 are computed on the held-out 20% test set that no model sees during training. This two-stage evaluation guards against overfitting and provides an unbiased estimate of out-of-sample predictive accuracy.

Model Comparison: Counterfactual Effects and Predictive Performance

	Linear OLS	GAM	Random Forest	Gradient Boosting
Increase postcode_lp by 0.10	19.68%	29.57%	24.38%	18.85%
Increase floorAreaSqM by 10%	5.99%	5.52%	4.09%	4.08%
Increase bedrooms by 1	-0.36%	-0.23%	2.69%	2.77%
Increase bathrooms by 1	12.02%	7.95%	7.69%	5.85%
Increase livingRooms by 1	4.55%	2.27%	3.67%	2.87%
Increase crime by 10%	0.38%	-0.33%	1.28%	2.85%
Halve distance to nearest tube	3.26%	3.33%	1.83%	3.14%
Leasehold compared to freehold	-9.28%	-10.75%	-10.58%	-17.73%
Root mean squared error (held-out)	0.2483	0.2042	0.1504	0.1310
R^2 (held-out)	0.8237	0.8808	0.9354	0.9510

Notes: Counterfactual effects represent average percentage changes in predicted prices under specified changes. Performance metrics are computed on held-out test data.

Model 1: Linear

The OLS specification serves as a natural baseline, imposing linear relationships between log prices and all predictors. This functional form assumes that the tenth square metre adds the same proportional value as the hundredth: an assumption we might reasonably doubt. Transforming amenity variables into logs helps capture diminishing returns but still constrains relationships to be monotonic.

OLS Regression Results with 5-Fold Cross-Validation

Variable	Coefficient	Robust SE	Sig.
Intercept	-13.375	0.190	***
Log Postcode Premium	1.794	0.014	***
Log Floor Area	0.612	0.005	***
Bedrooms	-0.005	0.002	***
Bathrooms	0.115	0.002	***
Living Rooms	0.044	0.002	***
Leasehold	-0.099	0.002	***
Log Crime	0.042	0.003	***
Log Nearest Shop	0.005	0.001	***
Log Nearest School	-0.008	0.001	***
Log Nearest Tube	-0.046	0.001	***
Log Tube Count	0.070	0.002	***
Log Nearest Hospital	-0.028	0.002	***
Log Nearest Park	-0.002	0.002	
Log Nearest Police	-0.031	0.002	***
Log Police Count	-0.043	0.005	***
Observations	108,632		

Cross-Validation Results (5-Fold):

Average RMSE = 0.2504

Average MAE = 0.1916

Average R^2 = 0.8220

Notes: Robust (HC1) standard errors reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

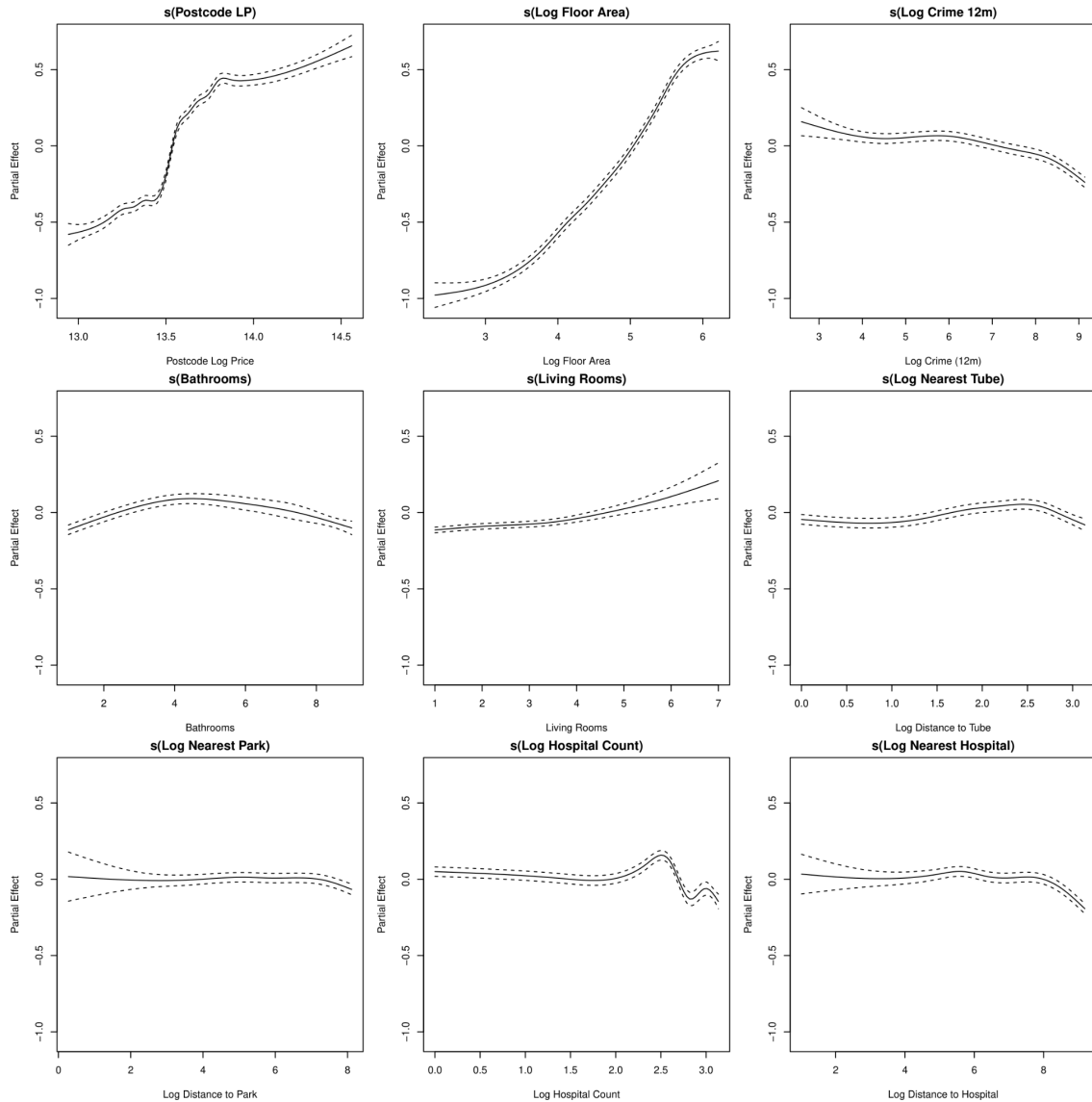
Despite these restrictions, the model achieves an adjusted R^2 of 0.82 and test RMSE of 0.25 on the log scale. Location premium dominates with a coefficient of 1.79, confirming the classic intuition that you're buying a neighbourhood as much as a building. Floor area contributes 0.61 per log unit, bathrooms add roughly 12% per unit, and leasehold tenure carries a 10% discount reflecting finite ownership horizons and ground rent concerns. Robust standard errors address heteroskedasticity, though the model cannot resolve omitted variable bias from unobserved neighbourhood characteristics.

The amenity coefficients appear to show very small effects and occasionally have a negative slope and so do bedrooms. This might be explained by the fact that we are controlling for floor area which explains some of the variation in price explained by an additional bedroom.

The linear structure is particularly restrictive for spatial and amenity variables, where marginal effects are unlikely to be constant across the distribution and interactions with property size and location cannot be accommodated.

Model 2: GAM

The GAM relaxes linearity by fitting smooth spline functions to each predictor, allowing the data to reveal curvature without imposing parametric forms. This flexibility proves valuable - adjusted R^2 rises to 0.88 and test RMSE falls to 0.20, suggesting hedonic price functions genuinely exhibit non-linearities that linear models miss. Importantly, these gains arise without introducing high-order interactions or discrete splits, preserving a notion of smooth hedonic pricing rather than piecewise segmentation.



The smooth terms reveal economically intuitive patterns: floor area effects might flatten at larger sizes as diminishing marginal utility sets in, bathroom premiums exhibit threshold effects, and transport

accessibility effects attenuate beyond practical commuting distances. The penalised splines with automatic smoothness selection guard against overfitting while capturing genuine curvature.

Model 3: Random Forest

Tree-based ensembles represent a fundamentally different philosophy: rather than specifying functional forms, the algorithm discovers structure through recursive partitioning. Each tree asks binary questions about features, and the forest aggregates 400 trees trained on bootstrap samples with random feature subsets.

This approach excels at capturing interactions - the effect of an extra bedroom might depend on floor area, or tube proximity might matter more in certain boroughs. The test R^2 of 0.94 and RMSE of 0.15 represent substantial gains over the GAM, suggesting interactions genuinely characterise London's market. Variable importance confirms the hierarchy: postcode location premium, floor area, bathrooms and bedrooms lead the rankings.

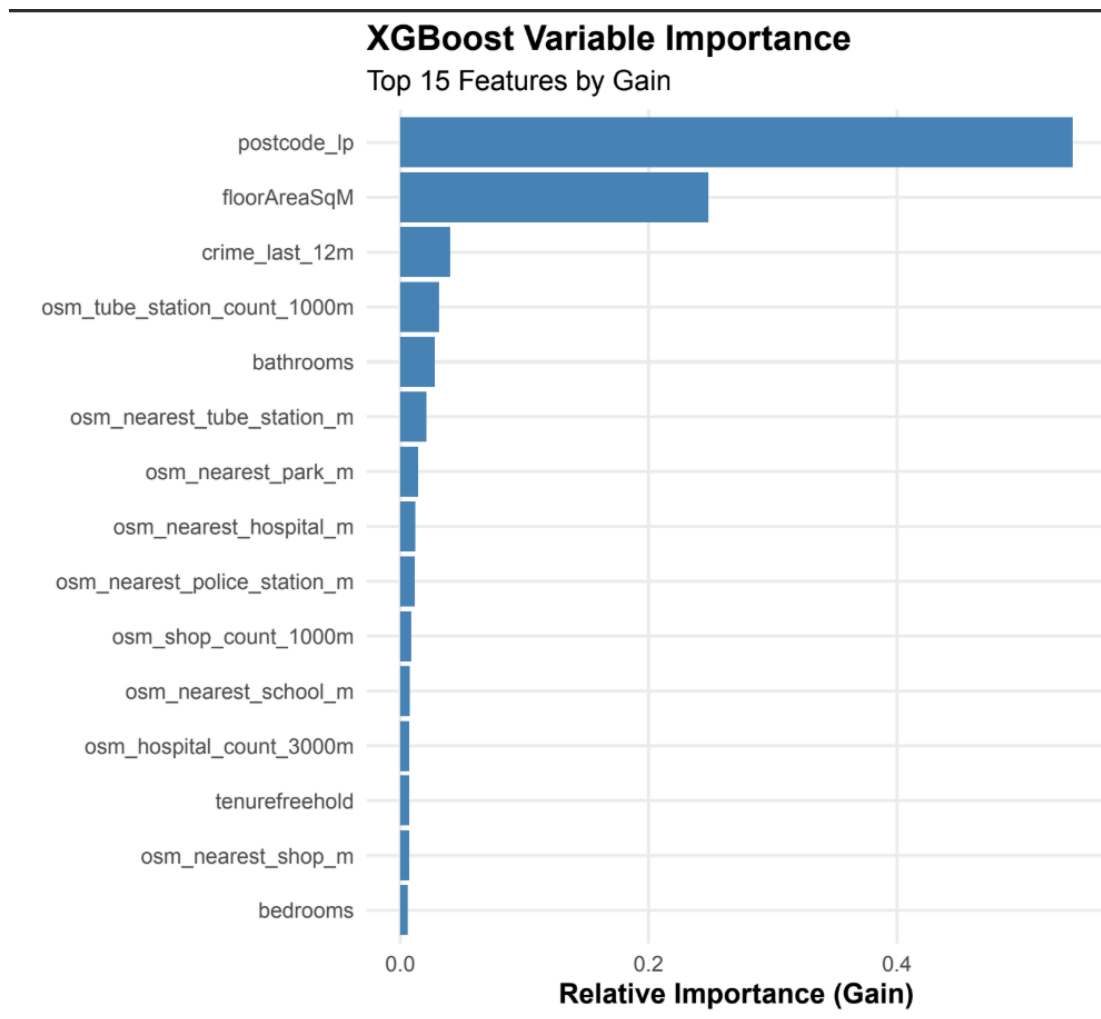
However, improved prediction comes at an interpretive cost. We lose the partial effect curves that make parametric models useful for policy analysis. The model becomes a black box: excellent for prediction but harder to interrogate for causal understanding.

Model 4: Gradient Boosting

Gradient boosting builds trees sequentially, with each new tree correcting errors from the ensemble so far. This iterative refinement achieves test R^2 of 0.95 and RMSE of 0.13, the best performance across all specifications. Training concluded at iteration 5,000 with training RMSE of 0.077 and test RMSE of 0.131, indicating well-controlled complexity without severe overfitting.

Feature importance based on gain shows sharp concentration: postcode location contributes 54% of total gain, floor area 25%, crime 4%, and tube station accessibility 3%. Transport variables and bathrooms collectively add around 6%. This suggests amenities operate as refinements once locational fundamentals are established.

The hyperparameters reflect the bias-variance tradeoff: a slow learning rate (0.05) allows many small corrections, tree depth and subsampling constrain complexity, and regularisation parameters prevent individual trees from overfitting.

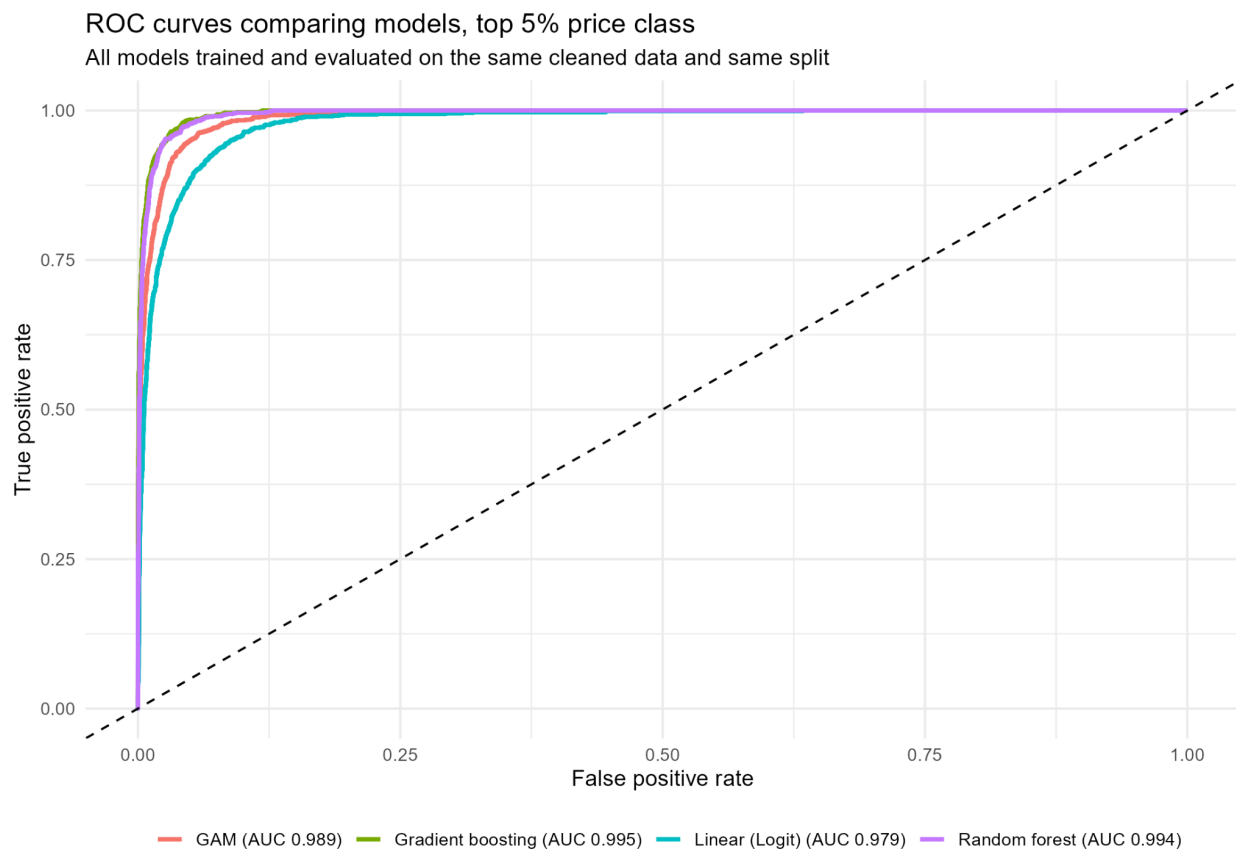


Summary of Model Results:

Overall, coefficient estimates align with economic intuition. Larger properties and freehold tenure command higher prices, while the postcode location premium emerges as one of the most influential predictors, confirming the dominant role of location in London housing markets. Amenity variables generally have the expected signs, though their magnitudes are moderated by regularisation.

Curiously, the rate of overall crime in the borough is positively associated with house prices, but this is not particularly surprising as our model takes into account overall crime which is dominated by petty crime. Petty crime is particularly prevalent in the more expensive boroughs of London as these are the areas targeted by pickpockets. Thus, the positive relation might be due to the extraneous factor of the behaviour of criminals to target inhabitants of these more expensive areas.

For the ROC analysis, prices were converted into a binary indicator equal to one if a property lies in the top 5% of log prices in the training sample. This tests our model on identifying the extremely high-end market. Models were evaluated on their ability to rank high-value properties using ROC curves and AUC.



Our analysis provided some counterintuitive insights. We found that the number of bathrooms had a much stronger effect on the price than the number of bedrooms. This could be overstated since our model holds square footage constant and bedrooms require more areas. However, this result could also be explained by

the fact that many flats in Central London have a high ratio of bedrooms to bathrooms and also due to it being easier to qualify a room as a bedroom whereas a bathroom requires plumbing.

Additionally, crime variable's positive coefficient in a few of our models likely reflects residual confounding with density and centrality rather than genuine preferences.

As a whole, the models demonstrate that feature engineering and regularised linear and non-linear methods can capture the main drivers of London house prices while remaining transparent and economically interpretable. Remaining errors are primarily attributable to unobserved heterogeneity and intangible factors rather than model misspecification.

Conclusion & Possible Future Improvements

Overall, the analysis shows that London house prices are driven primarily by location and scale, with postcode level effects dominating all models and floor area contributing large, but clearly non linear, price premia. Linear OLS provides a strong and interpretable baseline, but its constant marginal effects impose restrictive assumptions that are not fully supported by the data. Allowing for non linearities through GAMs and tree based methods reveals diminishing returns to size and amenities, and more complex interactions between location, property characteristics, and neighbourhood features. While machine learning models offer modest gains in predictive performance, they largely reinforce the same economic story uncovered by the simpler specifications. Taken together, the results suggest that functional form matters for interpretation rather than headline fit, and that careful modelling of location and scale is more important than adding ever more amenity variables.

Our models provide some valuation intuition, but improvements could be made with additional data or development time. The most important limitation is the absence of detailed property-level quality information, such as condition, refurbishment history, energy efficiency, floor level, and internal layout. These factors explain a large share of price variation within postcodes and their inclusion would likely reduce residual error substantially.

Additionally, we did not apply formal variable selection procedures such as LASSO to the linear specification, meaning some weak predictors may be retained unnecessarily. Moreover, while GAM and XGBoost incorporate built-in regularisation, we did not systematically tune the penalisation strength or conduct sensitivity analysis on these parameters. A more rigorous approach would involve cross-validated

selection of regularisation hyperparameters across all model classes, together with sensitivity analysis to assess robustness to alternative penalty strengths.

Furthermore, we realise that aggregating crime data at the borough level is a simplification and we would have liked to use more localised data. Our data also likely underrepresents council homes since these are only available to buyers through the Right to Buy scheme and not present in our dataset.

Our findings align with a substantial body of prior research on urban housing markets. Gibbons and Machin (2008) review the literature on hedonic house price determinants and emphasise that housing prices exhibit highly localised variation that cannot be explained solely by physical dwelling attributes: a pattern strongly confirmed by our postcode-level fixed effects. Their survey highlights transport accessibility, school quality, and crime as key policy-relevant amenities capitalised into prices, though they note serious empirical challenges in isolating causal effects from correlated neighbourhood characteristics. Like earlier studies, we find that careful treatment of location dominates model performance, while the marginal contribution of additional amenity variables diminishes rapidly once spatial heterogeneity is adequately controlled.

With more time, further gains could also be achieved through richer spatial modelling and model refinement. Finer geographic identifiers or spatial methods could better capture street-level variation, while more extensive hyperparameter tuning and segment-specific modelling could improve performance at the margin. These enhancements would not alter the core findings, but would likely push predictive accuracy closer to its feasible upper bound.

In terms of relevance for policies, our findings suggest that new housing on peripheral green belt land (as suggested by Sadiq Khan) would command substantially lower prices than equivalent properties in established inner boroughs, reflecting the powerful capitalisation of locational reputation and transport accessibility. Policymakers should anticipate that affordability gains from green belt development may be partially offset by reduced amenity valuations in these locations. Ultimately, this analysis confirms what London buyers intuit: you pay overwhelmingly for where you live, and only secondarily for what you live in.

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Gibbons, S. and Machin, S. (2008) 'Valuing school quality, better transport, and lower crime: evidence from house prices', *Oxford Review of Economic Policy*, 24(1), pp. 99–119. Available at:

<https://academic.oup.com/oxrep/article-abstract/24/1/99/480352>

Datasets

<https://www.kaggle.com/datasets/jakewright/house-price-data>

https://github.com/LingruFeng/GIS_assessment/raw/main/London_Boroughs.gpkg

<https://www.ons.gov.uk/datasets/cpih01/editions/time-series/versions/66>

OpenStreetMap API

AI Declaration:

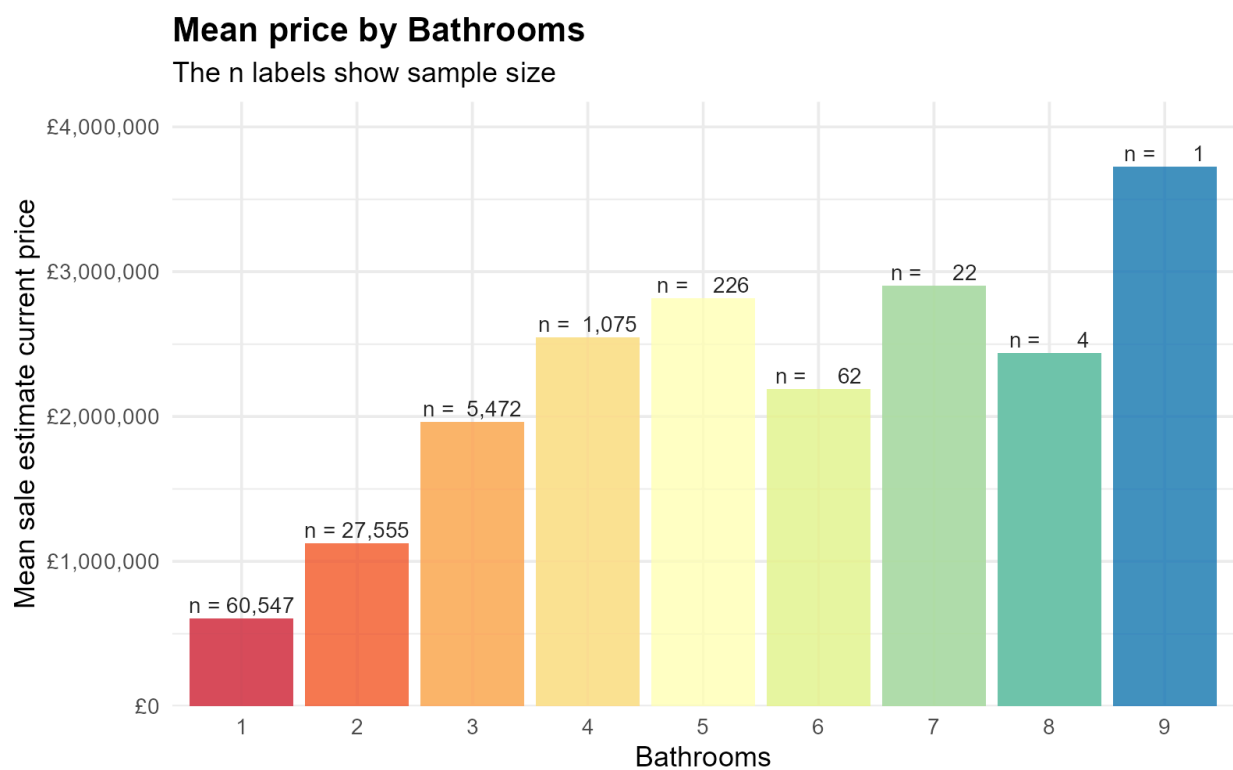
We used AI primarily to debug our code and to understand our errors better. Particular instances include when we were building our random forest model but it was taking very long to run so we asked ChatGPT for an efficiency improvement and it recommended that we use the Ranger package. We also used it to help us find a github source where we could input our observations and find which London Borough they were in so we could find the corresponding crime data.

Finally, we used it to improve our grammar in our report and for idea generation for graphs.

Appendix 1:

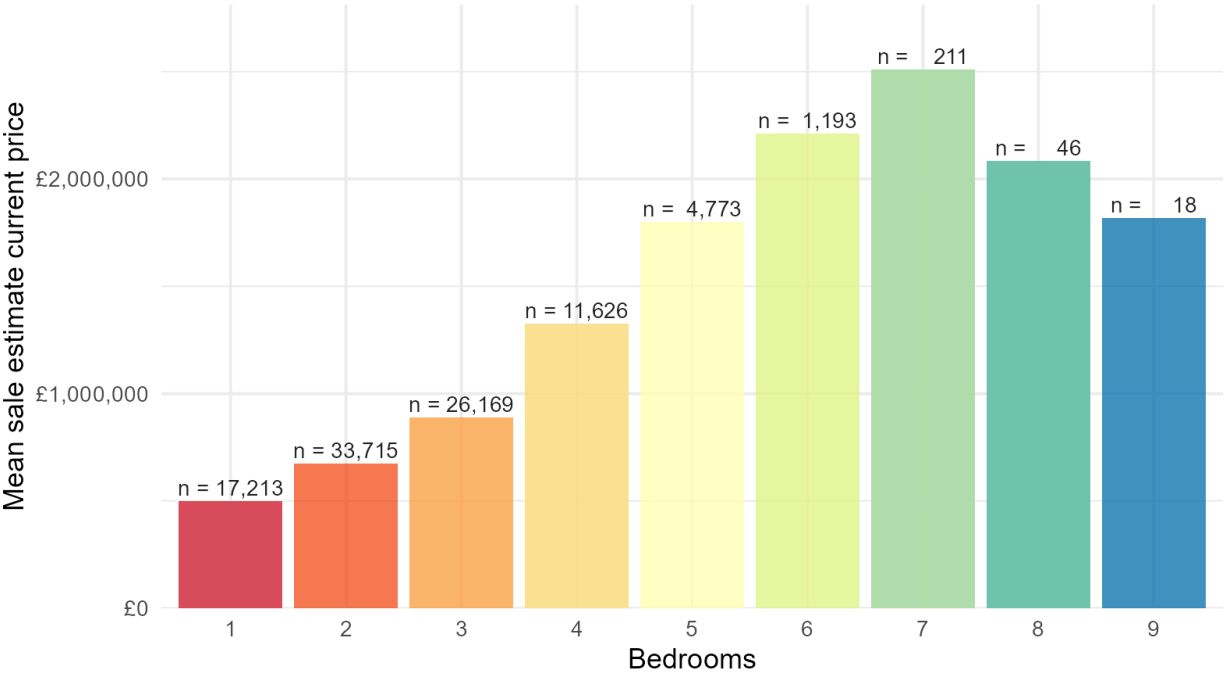
We used Dataset 1 (without imputing missing values) for the graphs based on initial data exploration like below.

The results support our finding that bathrooms have a bigger effect on sale prices than bedrooms since as we can see below that the majority of houses have between 1-3 bathrooms whereas there are still many houses with higher number of bedrooms.



Mean price by Bedrooms

The n labels show sample size



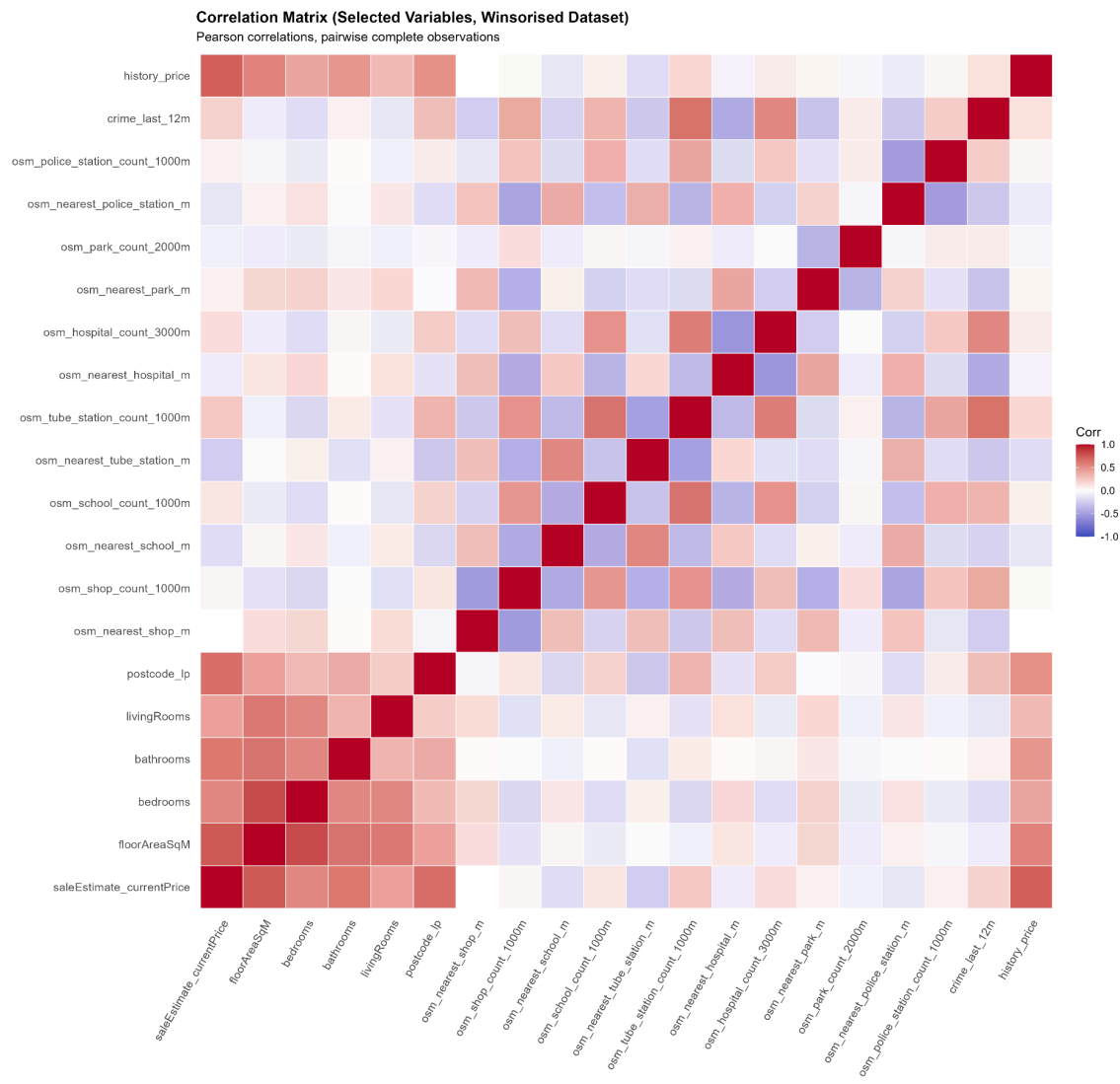
Appendix 2:

The data below looks at the sales price current estimate and controls for the other attributes like bedrooms and bathrooms, and it finds the % premium associated with an apartment if it is located in a certain post code compared to the median of the dataset. It follows that the highest premium houses are in Central London in areas like Mayfair and Belgravia.

Top Postcode District Premiums

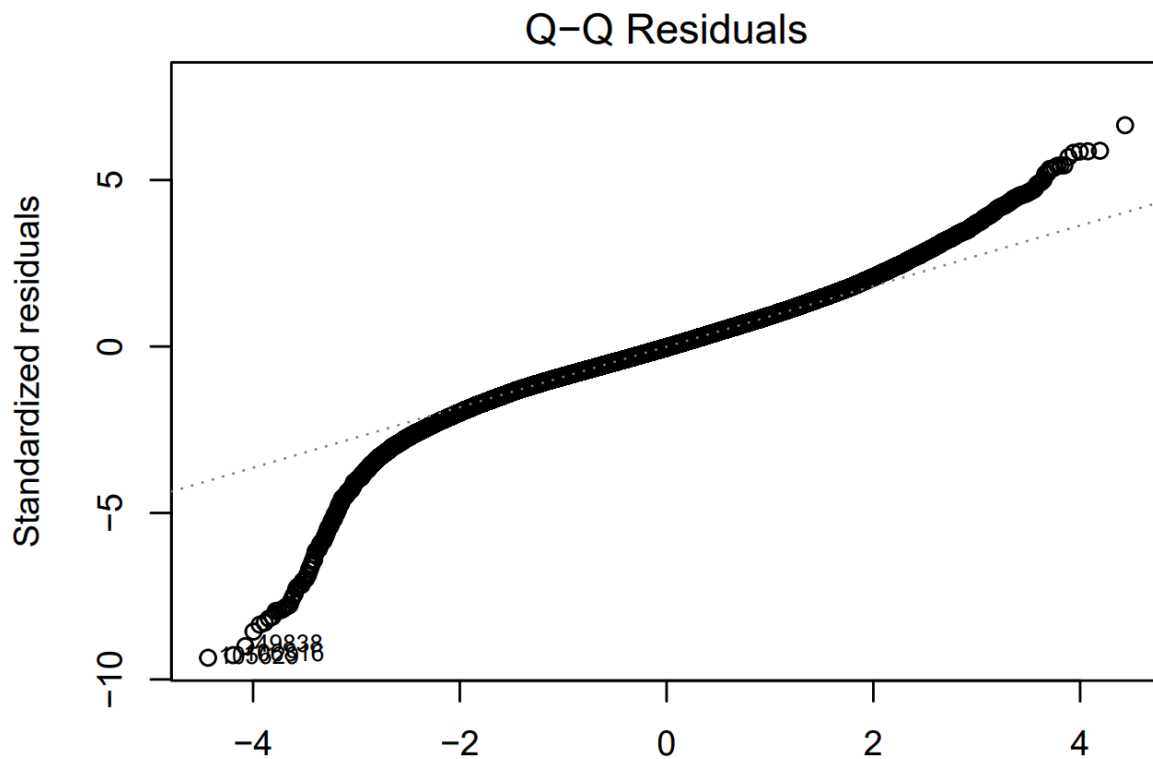
Postcode District	Premium (%)
W1S	265.74
SW1X	248.52
W1J	229.30
SW7I	226.80
SW32	209.61
WC2R	200.78
W1K	197.96
SW72	174.59
SW34	171.55
W1B	169.57
SW31	166.96
SW1Y	166.86
SW1A	163.39
W1T	160.65
W87	160.56
W1G	159.73
W85	156.53
W1U	150.49
SW73	147.14
WC2E	144.71

Appendix 3:



Appendix 4:

The Q-Q plot reveals substantial departures from normality, particularly in the tails, while the central portion follows the theoretical line reasonably well. These heavy tails suggest the presence of extreme observations that the model struggles to accommodate, indicating potential heteroskedasticity, which was addressed through the use of robust standard errors to ensure reliable inference. However, with over 100,000 observations in the dataset, the Central Limit Theorem ensures that coefficient estimates and hypothesis tests remain asymptotically valid despite non-normal residuals.



Appendix 5:

The residuals exhibit a clear funnel pattern with increasing dispersion at higher fitted values, indicating heteroskedasticity. Accordingly, all inference relies on heteroskedasticity-robust and borough-clustered standard errors to ensure validity under arbitrary variance heterogeneity and within-area dependence. The absence of strong curvature in the residual pattern suggests the linear specification adequately captures average relationships in the data, though our non-linear models better accommodate the heterogeneity observed at the upper tail of the price distribution.

